

Huynh Le Dang Khoa

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Summary

As an Honors B.Eng. CS student at HCMUT, I contribute to collaborative research at **ARENA Lab** and **SMART** (Singapore-MIT Alliance for Research and Technology) on computer vision, reinforcement learning, and vision-language models for perception and robot learning.

- Contribute to collaborative projects spanning referring segmentation (**HORSE**), Isaac Lab synthetic data (**SynTable**), and behavior-aware social navigation (**BLINC** on Arena-Rosnav / ROS 2).
- Specialize in geometry-grounded vision-language grounding, zero-shot / synthetic-data segmentation, multi-modal perception (detection, pose, depth), and GPU-efficient inference profiling.

Education

Ho Chi Minh City University of Technology (HCMUT – VNU-HCM)
B.Eng. Computer Science & Engineering — Honors Program (Top 5% of Faculty)

Ho Chi Minh City, Vietnam
2022 – 2026

- **GPA:** 3.6 / 4.0 · [Academic Transcript](#)

Experience

Eye Code Tech — Arena Lab (Berlin & Singapore)

Research Collaborator — Vision AI, Simulation & Robot Learning

Singapore (Remote)

Feb 2026 – Present

- **Collaborative Research:** Contribute to joint projects at **ARENA Lab** and **SMART** (Singapore-MIT Alliance for Research and Technology) on vision, simulation, and robot learning.
- **Synthetic Data (Isaac Lab):** Built automated **synthetic data** pipelines in **Isaac Lab** for **zero-shot** segmentation when real labels are scarce (**SynTable**).
- **Multi-Modal Perception:** Train/evaluate **object detection**, 6-DoF pose, depth, and segmentation models in **PyTorch**; ID/OOD and sim-to-real benchmarks combining **2D**, **3D**, and scene context.
- **World Models & ROS:** Experiment with latent **world models** (e.g., **LE-WM**-style) for predictive scene understanding; contribute to **Arena-RosNav** — mobile navigation in Gazebo/**Isaac Lab** with pedestrian agents and **ROS 2 (Jazzy)** policies; [GitHub].
- **HORSE & Scale:** Co-authored **HORSE**, a lightweight geometry-grounded relational reranking adapter for frozen **SAM3** referring segmentation (AAAI'27 under review); train and ablate at scale on **8× NVIDIA H100**; [OpenReview].

HPC Lab (Research Group), HCMUT

Research Assistant — GPU Inference, Scheduling & Profiling

Ho Chi Minh City, Vietnam

Jan 2025 – May 2026

- **High-Performance Inference (Thesis):** Led adaptive scheduling for mixed ML workloads: packaged **YOLO/ResNet**, profiled **latency**/throughput/memory on **NVIDIA RTX 5090**, and tuned batching under SLA targets for real-time serving.
- **Systems & Evaluation:** Designed reproducible benchmarks vs. strong baselines; analyzed accuracy-latency trade-offs for GPU-accelerated inference pipelines.

Projects

SynTable — Isaac Lab Synthetic Data (with Do Van Minh) · Synthetic data generation in **NVIDIA Isaac Lab** for training and evaluating **segmentation** models in **zero-shot** settings — producing labeled cluttered scenes when real annotated data is limited.

- ▷ **Contribution:** Automated scene generation and labeling pipelines to support robust segmentation without heavy reliance on real-world annotations.
- ▷ **Stack:** **Isaac Lab**, **Python**, **PyTorch**, synthetic-to-real evaluation.

HORSE — AAAI 2027 Submission (Co-Author) · Lightweight **geometry-grounded relational reranking** adapter for referring segmentation: reranks confidence-filtered **SAM3** candidates via box geometry, self-attention, and frozen CLIP text — no mask-generator retraining; [OpenReview].

- ▷ **Results:** Improves SAM3 on all nine splits — RefCOCO+ TestB **43.68**→**54.64** (+10.96), RefCOCO+ Val **+9.10**, RefCOCO TestB **+8.68**; largest gains on relation-intensive / same-category expressions. Stack: **Python**, **PyTorch**, **SAM3**, **CLIP**.

Adaptive ML Inference Server — Graduation Thesis, Score 9.84/10 (Highest in Cohort) · Adaptive scheduling for concurrent inference under bursty traffic; packaged **YOLO/ResNet**, profiled on **RTX 5090**; ~20% higher throughput and ~80% fewer SLA violations vs. A* baselines.

- ▷ **Stack:** **Python**, **ONNX Runtime**, GPU latency profiling, **Linux**.

Publications & Achievements

Publications: *HORSE: Geometry-Grounded Relational Reranking for Referring Expression Segmentation*. Co-Author, AAAI 2027 (under review). [OpenReview](#) · *An Intelligent Virtual Machine Consolidation Approach...* First Author, 19th ACOMP (2025). DOI

Skills: **Python**, **C++**; **Linux**; **PyTorch**, **OpenCV**, **CUDA**; **Isaac Sim/Isaac Lab**, **ROS 2**; **YOLO**, **ResNet**, **SAM3**; **ONNX** / **TensorRT**-ready packaging; GPU profiling (latency, memory); concurrency & systems fundamentals; **Git**.

Awards: Most Outstanding Intern — RAG AI Agent (Dien Toan Group); 2nd Place, Innostart Entrepreneurship Competition.

Languages: English (Professional Working Proficiency); Vietnamese (Native).